

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	SECTION:	CPE21S1
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Prelim Laboratory Exam SET C

1. PROCEDURE OUTPUT

The Palindrome Number Problem

Problem Definition:



A palindrome number is a number that remains the same when digits are reversed. For example, the number 12321 is a palindrome number, but 1451 is not a palindrome number. Given a positive integer, the task is to check whether the number is palindrome or not. We want to use data structures to check whether a given string of input conforms to the given description of a palindrome. The user is then prompted whether the given input is a palindrome or not.

Your task is to use two data structures for your solution:

Stack
Queue

But, they must have different internal representations, either:

Array-based
Linked List

Example: If your stack is array-based, your queue must be implemented using a linked list (and vice-versa).

You must then use the values in the given table below to validate your code.

Palindrome?	Output (Console)	Remark
10201		
1003003001		
88		
0		
202		

Prelim_Exam.cpp

```
#include <iostream>
#include "ArrQueue.h"
#include "LLStack.h"

bool isPalindrome(std::string text){
    ArrayQueue q;
    LinkedStack s;
    for(int i=0;i<text.length();i++){

        q.enqueue(text[i]); //Queue object named q.

        s.push(text[i]); //Stack object named s.
    }

    while(!s.isEmpty())
    {
        char a = q.dequeue();

        char b=s.pop();

        if(a!=b) return false;
    }
    return true;
}

int main(){
    std::string input;
    std::cout<<"Enter a number: ";
    std::cin>>input;

    if(isPalindrome(input))
        std::cout<<input<<" is a palindrome number."<<std::endl;
    else
        std::cout << input << " is not a palindrome number." <<
```

```
std::endl;  
  
return 0;  
}
```

LinkedStack.h

```
#ifndef LINKEDSTACK_H  
#define LINKEDSTACK_H  
#include <iostream>  
  
struct Node{  
char data; Node* next;  
    Node(char v){data=v; next=NULL;}  
};  
  
class LinkedStack {  
Node* top;  
    int count;  
public:  
    LinkedStack(){ top=NULL; count=0; }  
    ~LinkedStack(){ while(!isEmpty()) pop(); }  
  
bool isEmpty(){  
    return top==NULL;  
}  
  
void push(char value)  
{  
    Node* n = new Node(value);  
    n->next=top;  
    top=n; count++;  
}  
  
char pop(){  
if(isEmpty()){std::cout<<"Stack underflow!"<<std::endl; return '\0';}
```

```
Node* t=top; char v=t->data;
    top=top->next;

    delete t;
count--;
    return v; }

    int size() { return count; }
};

#endif
```

ArrayQueue.h

```
#ifndef ARRAYQUEUE_H
#define ARRAYQUEUE_H
#include <iostream>

class ArrayQueue{
    private:
        int data[10];
        int front;
        int rear;
        int count;

    public:
        ArrayQueue() {
            front=0;
            rear=-1;
            count = 0;}

        bool isEmpty(){ return count==0; }
        bool isFull() {
            return count==10; }

        void enqueue(char value){
```

```
if(isFull()) {std::cout<<"Queue overflow!"<<std::endl; return;}
    rear=(rear+1)%50;
    data[rear]=value;
count++;
}

char dequeue()
{
    if(isEmpty()) { std:: cout << "Queue underflow!" << std::endl;
    return '\0'; }
    char value=data[front];
        front=(front+1)%50;
    count--;
return value;
}

int size(){return count;}
};

#endif
```

OUTPUT

```
\CPE 010-CPE21S1\exam_prep(prelim)\\" ; if ($?) { g++ Prelim_exam.cpp -o Prelim_exam } ; if
Enter a number: 10201
10201 is a palindrome number.
PS C:\Users\Benj Federic De Leon\Documents\CPE21S1\CPE 010-CPE21S1\exam_prep(prelim)> cd "c
\CPE 010-CPE21S1\exam_prep(prelim)\\" ; if ($?) { g++ Prelim_exam.cpp -o Prelim_exam } ; if
Enter a number: 1003003001
1003003001 is a palindrome number.
PS C:\Users\Benj Federic De Leon\Documents\CPE21S1\CPE 010-CPE21S1\exam_prep(prelim)> cd "c
\CPE 010-CPE21S1\exam_prep(prelim)\\" ; if ($?) { g++ Prelim_exam.cpp -o Prelim_exam } ; if
Enter a number: 88
88 is a palindrome number.
PS C:\Users\Benj Federic De Leon\Documents\CPE21S1\CPE 010-CPE21S1\exam_prep(prelim)> cd "c
\CPE 010-CPE21S1\exam_prep(prelim)\\" ; if ($?) { g++ Prelim_exam.cpp -o Prelim_exam } ; if
Enter a number: 0
0 is a palindrome number.
PS C:\Users\Benj Federic De Leon\Documents\CPE21S1\CPE 010-CPE21S1\exam_prep(prelim)> cd "c
\CPE 010-CPE21S1\exam_prep(prelim)\\" ; if ($?) { g++ Prelim_exam.cpp -o Prelim_exam } ; if
Enter a number: 202
202 is a palindrome number.
PS C:\Users\Benj Federic De Leon\Documents\CPE21S1\CPE 010-CPE21S1\exam_prep(prelim)> █
```

Code analysis:

The code uses a Queue and Stack to store each digit of the input number. The Queue gets the digits in their original order using enqueue(), while the Stack gets them in reverse order using push(). The program then compares the digits using dequeue() and pop(); if they are all the same, the number is a palindrome, otherwise it is not.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

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